

UNIT IV :- NETWORK TOPOLOGIES & NETWORKING DEVICES

* Network Topology :-

- A network topology is a physical and logical arrangement of nodes & connections in a network.
- nodes include - switches, routers & software with switch & router features.

• Categories of network topology :-

1) Physical topology :-

Defines how nodes of network are physically connected.

2) Logical topology :-

How data is transmitted between nodes.

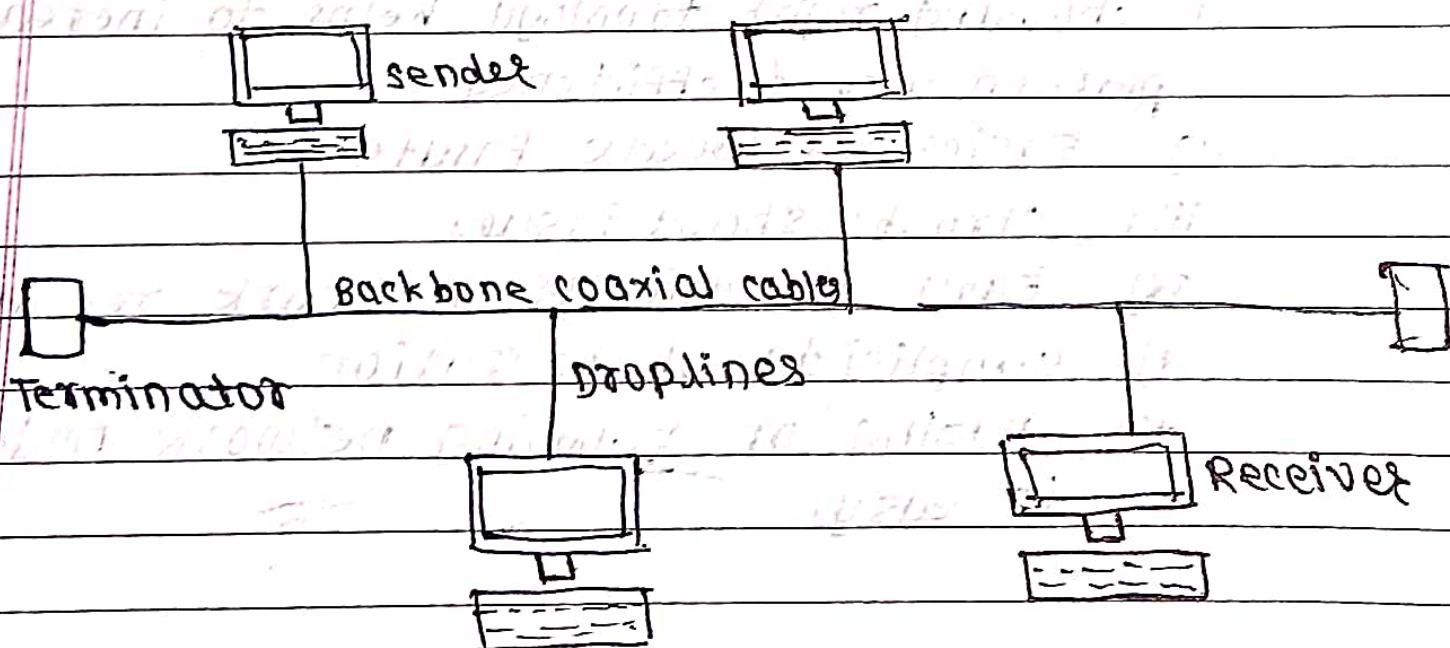
• Benefits :-

- i) choosing right topology helps to increase performance & efficiency
- ii) Easier to locate faults
- iii) Troubleshoot issues
- iv) Easy to allocate network resources
- v) simplicity of operation
- vi) Adding or removing network nodes is easy.

1]

Bus Topology :-

- i) Every computer & network device is connected to a single cable.
- ii] It is bidirectional.
- iii] In this, if backbone fails the topology crashes.
- iv] Devices share responsibility for getting data from one point to another.
- v] When device sends msg, it is broadcast down on cable in both directions. Terminators at end of cable prevent the signal from reflecting back to the sender.
- vi] All devices on cable constantly monitor for msg meant to them. When device detects msg meant for it, it reads msg for cable & other devices will ignore it.
- vii] Drop line - is connection between devices & cable.



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Advantages:-

- i) Easy to install
- ii) Less cabling
- iii) Less expensive

Disadvantages:-

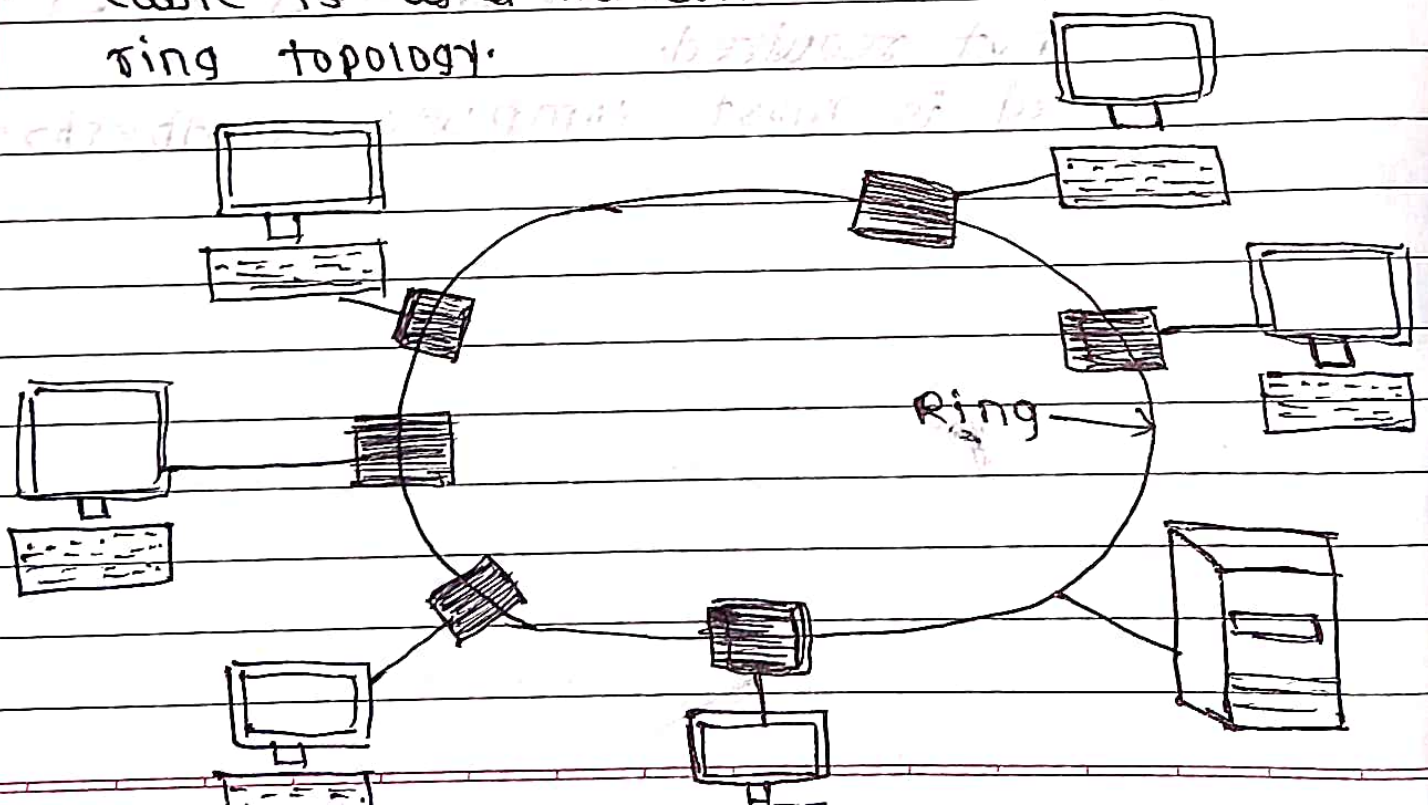
- i) Difficult reconfiguration & fault isolation
- ii) Difficult to add new device
- iii) degradation in quality can cause due to signal reflection ^{at} top
- iv) IF there is any fault in backbone, it can stop all transmission.
- v) security low

Applications:-

- used when small, cheap & temporary network is needed.
- when high data transfer speed is not required.
- used in most computer motherboard

2] Ring Topology:-

- i) Each device is connected with two devices on either side.
- ii) signal is passed From device to device until it reaches destination & each device have repeater
- iii) when one device received signals instead of intended another device, its repeater then regenerates the data & passes them along.
- iv) To add or delete device requires changing only two connections.
- v) Devices are connected in ring structure.
- vi) packets travel in only ~~two~~ one direction
- vii) Depending on network card used in each computer, RJ-45 network cable or coaxial cable is used to connect computers in ring topology.



Advantages:-

- i) Easy to install
- ii) Easy to reconfigure
- iii) Fault identification is easy

Disadvantages:-

- i) Unidirectional traffic
- ii) Break in single ring can break entire network

Applications:-

- i) Offices
- ii) School campuses
- iii) Used where simple network is needed

3] STAR TOPOLOGY:-

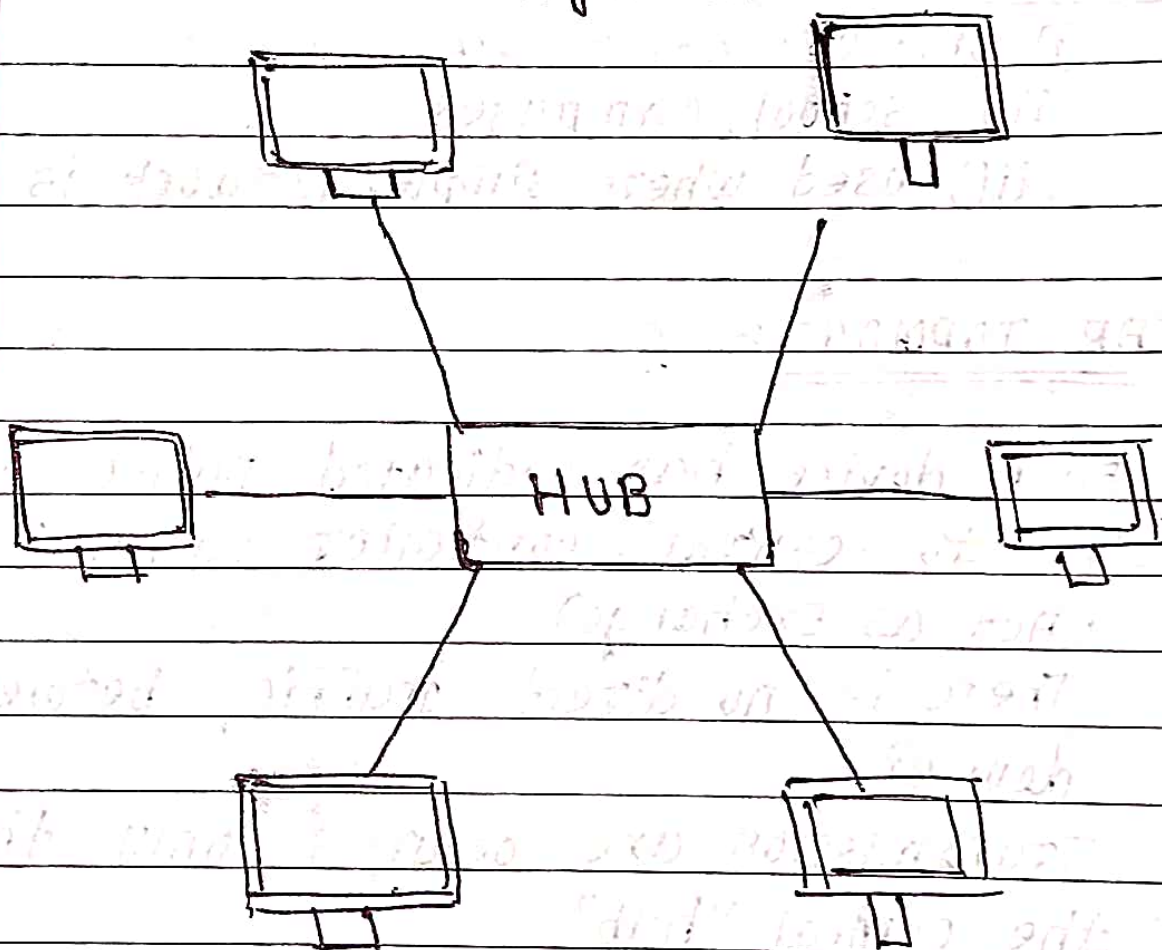
- i) Each device has dedicated point-to-point link to central controller called "Hub" (Act as Exchange)
- ii) There is no direct traffic between devices
- iii) Transmission are occurred only through the central "hub"
- iv) Peripheral devices act as clients & hub acts as servers
- v) RJ 45 or coaxial cable is used according to network card installed in each computer

vi) Central point maybe passive, active or intelligent hub.

passive hub - connects arms of star but no signal is regenerated

Active hub - It regenerates signals

Intelligent hub - not only regenerate signals but also perform activities such as intelligent path selection & network management.



- Advantages :-

- i) Less expensive than mesh
- ii) Installation & configuration easy
- iii) Less cabling than mesh
- iv) Robustness (If one link fails, only that link is affected. All other links remain active)
- v) Easy to Fault identification & to remove parts.

- Disadvantages :-

- i) It is large compared to other topologies
- ii) When hub goes down, whole system is dead

- Applications :-

- i) Used in LANs
- ii) High speed LAN
- iii) small organizations

Q) MESH TOPOLOGY :-

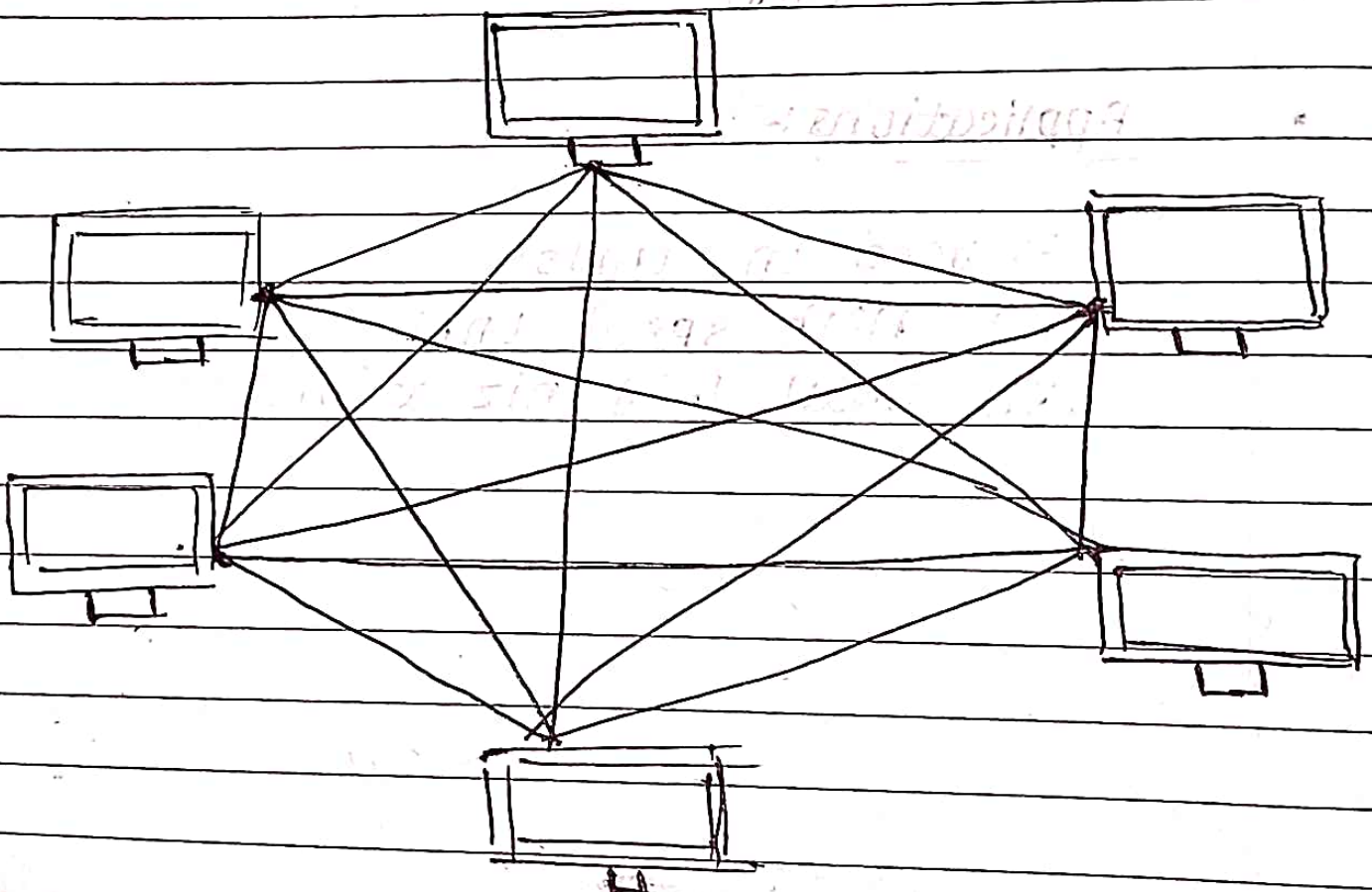
i) Each comp & network device is inter-connected with one another, allowing most transmissions to be distributed, even if one connections go down.

ii) Not commonly used, coz it is difficult & expensive ^{for comp networks}

iii) commonly used for wireless networks

iv) Connected nodes can be computers, switches, hubs or any other devices.

v) If break occurs in segment of cable, the traffic load of network is redistributed between all nodes, which maintains availability of network.



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• Advantages:-

- i) They use dedicated links, so traffic problem can be avoided.
- ii) It is robust.
- iii) Gives privacy & security.
- iv) Fault identification & Fault isolation are easy.

• Disadvantages:-

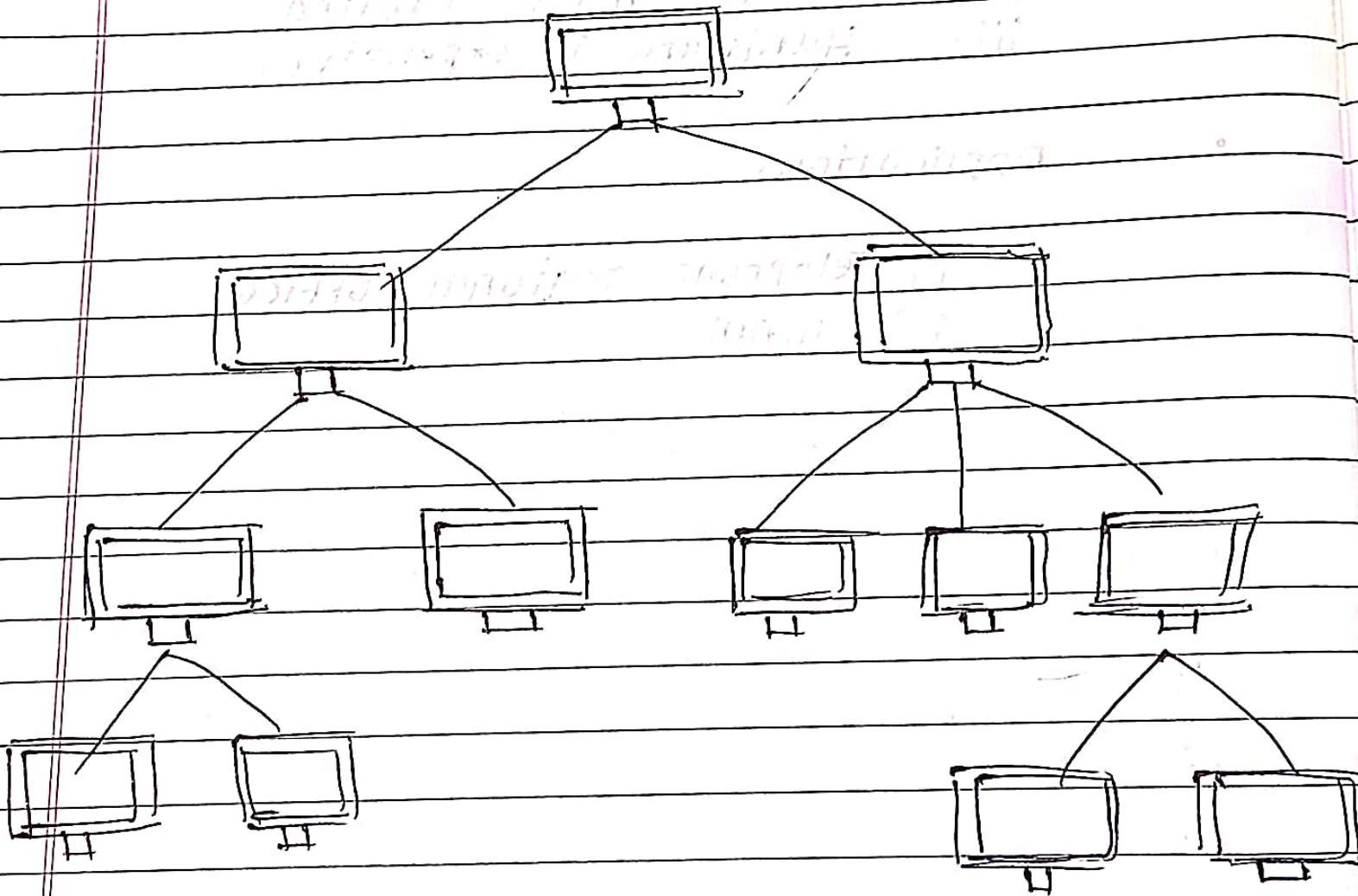
- i) Expensive.
- ii) Requires more cabling.
- iii) Hardware is expensive.

• Applications:-

- i) Telephone regional office.
- ii) WAN.

5) TREE TOPOLOGY:-

- i) also called as star bus topology
- ii) most common network setups
- iii) It connects multiple star networks to other star networks.
- iv) All elements are arranged like branches of a tree.
- v) Every node in this architecture is connected one to one in a hierarchy level, with each neighbouring node on its lower level.



• Advantages:-

- i] hierarchical as well as central data arrangement
- ii] Easy maintenance
- iii] Easy Fault detection
- iv] Highly secure
- v] Used in WAN
- vi] Reliable

• Disadvantages:-

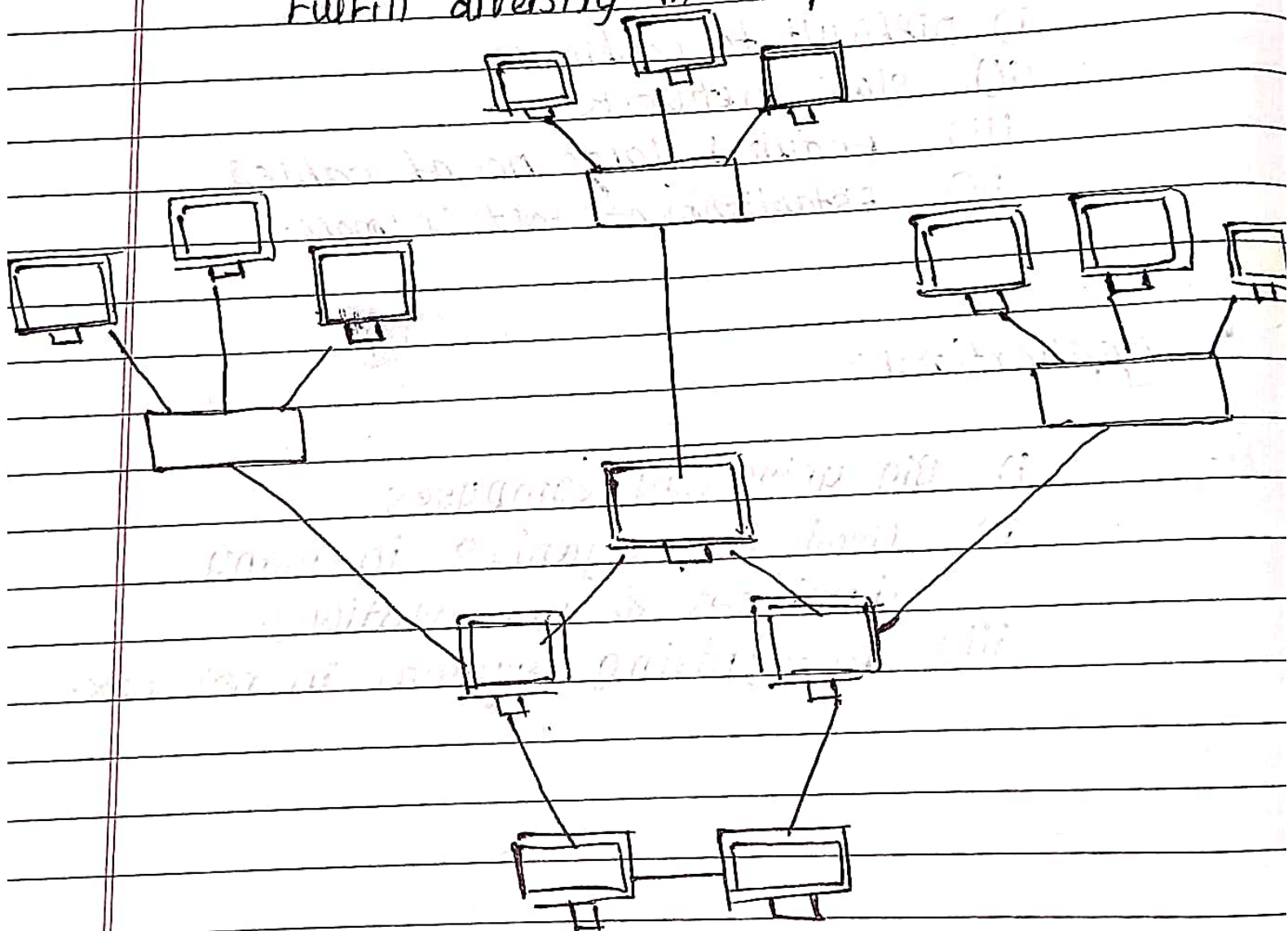
- i) Difficult to configure
- ii) slow network
- iii) Requires large no. of cables
- iv) Establishment cost is more.

• Applications:-

- i) Big university campuses
- ii) Used for organize in many databases & workstations.
- iii) Recognising system in network.

6] HYBRID TOPOLOGY :-

- i) A network which contain all type of physical structure & connected under a single backbone channel.
- ii) It is combination of two or more network topologies.
- iii) It increase Fault tolerance, & allows new basic topologies to be added or removed easily.
- iv) It is more useful when u need to fulfill diversity in computer network.



• Advantages:-

- i) Reliable
- ii) Effective
- iii) Scalable
- iv) Flexible

• Disadvantages:-

- i) Complexity

- ii) Expensive

• Applications:-

- i) Financial & banking sector

- ii) Automated industries

- iii) Multi-national companies

- iv) Research organizations

- v) Educational institutions

network devices- piece of hardware used to handle electrical/digital connections and perform their unique roles efficiently.

4.2] NETWORK CONTROL / CONNECTING DEVICES :-

1] Hub :-

- i) multi port repeater
- ii) connects multiwires coming from different branches. for ex- connector in star topology. which connects different stations.
- iii) Hub can't filter data, so data packets are sent to all connected devices.
- iv) Also, hubs don't have intelligence to find out the best path for data packets which leads to inefficiencies & wastage.

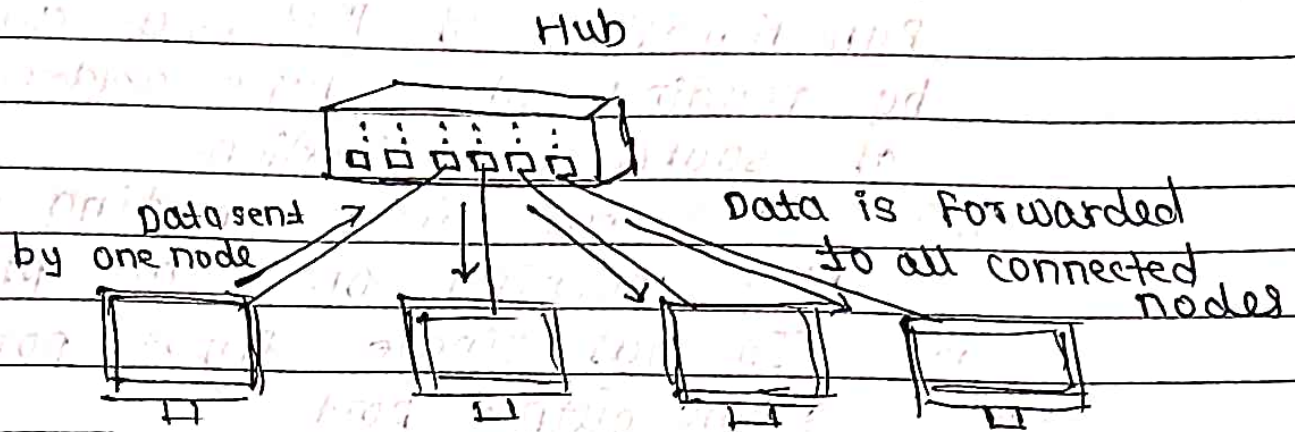
• Types of hub :-

- 1) Active hub :-
- i) have their own power supply
 - ii) serves as repeater & wiring centre
 - iii) used to extend maximum distance between nodes

- 2) passive hub :-
- i) ~~can~~ collect wiring from nodes & power supply from active hub.
 - ii) relay signal on network without cleaning & boosting them.
 - iii) can't be used to extend distance between nodes

3] Intelligent hub -

works like active hub & includes remote management capabilities.



working of hub

2] Repeater :-

i] operates at physical layer

ii] Its job is to regenerate the signal over same network before signal becomes too weak or corrupted to extend the length to which signal can be transmitted over the same network

iii] They don't amplify signal.

iv] when signal becomes weak, they copy it bit by bit & regenerate it as its star topology connectors connecting it original strength.

v] It is a 2 port device.

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4] Bridges :-

- i) Operates at the data link layer.
- ii) It is repeater, with add on the functionality of filtering content by reading the MAC addresses of source & destination.
- iii) Used for interconnecting two LANs working on same protocol.
- iv) It has single input port & single output port.

4] Switches :-

- i) Multiport bridge
- ii) Data link layer device
- iii) It can perform error checking before forwarding data.
- iv) It doesn't forward packets having errors.
- v) It divides collision domain of hosts but broadcast domain remains same.

5] Routers :-

- i) It is device like switch
- ii) Routes data packets based on their IP addresses.
- iii) Network layer device
- iv) Normally connect LANs & WANs and have a dynamically updating

routing table based on which they make decisions on routing data packets.

v] The router divides broadcast domains of hosts connected through it.

6] Gateway :-

i] Passage to connect two networks that may work upon different networking models.

ii] They work as messenger agents that ~~no~~ take data from one system, interpret it & transfer it to another system.

iii] Gateways are also called protocol converters and can operate at any network layer.

iv] Gateways are generally more complex than switches or routers.

v] Gateway is also called a protocol converter.

7] modem :-

i] modem stands for modulation Demodulation

ii] modem converts digital data signals into analogue data signals.

iii] They can be installed within the computer in a development slot applicable for it.

iv] Modems have high uploading

4] communication rates.

v] They enable high speed downstream data transfers by digitally encoding all downstream data.

8] Firewall :-

i] Firewall is a network security device.

ii] It manages all incoming & outgoing traffic and based on a defined set of security rules it accepts, rejects or drops that specific traffic.

iii] Firewall establishes a barrier between secured internal networks and outside untrusted network, such as the Internet.

iv] It is network security device.